RAJALAKSHMI ENGINEERING COLLEGE

Mridula S 251701029-2025-CSD

M2

Started on	Wednesday, 30 September 2026, 9:28 PM
State	Finished
Completed on	Wednesday, 30 September 2026, 9:29 PM
Time taken	1 min 14 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Problem Statement

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

// Function to find the first occurrence of 0 using divide and conquer
int findFirstZero(int arr[], int low, int high, int size) {
    if (high >= low) {
        int mid = low + (high - low) / 2;

        // Check if this is the first zero
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0)
            return mid;

        // If element at mid is 1, search in right half
        if (arr[mid] == 1)
            return findFirstZero(arr, mid + 1, high, size);

        // Else search in left half
        return findFirstZero(arr, low, mid - 1, size);
    }
    return -1; // No zero found
}

int main() {
    int m;
    scanf("%d", &m);

    int arr[m];
    for (int i = 0; i < m; i++) {
        scanf("%d", &arr[i]);
    }

    int firstZeroIndex = findFirstZero(arr, 0, m - 1, m);

    if (firstZeroIndex == -1)
        printf("0\n"); // No zero present
    else
        printf("%d\n", m - firstZeroIndex); // Count of zeros

    return 0;
}

```

	Input	Expected	Got	
✔	5 1 1 1 0 0	2	2	✔

	Input	Expected	Got	
✔	10 1 1 1 1 1 1 1 1 1 1	0	0	✔
✔	8 0 0 0 0 0 0 0 0	8	8	✔
✔	17 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0	2	2	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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